Ex 2.2.16: the in-context grammar pilot

> The first experiment on the grammar from the D2.2 pivot: the model infers the op from a few solved examples, with no word to name it. We trained the control at three conditions of example count and replacement rate, and anchored `difference` under the whole-line label and its variants. Every control got a little under halfway from the floor to the Bayes ceiling (skill scores 0.38 to 0.46), well short of the pass line, and so did the d128-L4 control. So the pilot stops at its first rule: the grammar and recipe are reworked before anything is anchored, and the other rules are not scored. Ex-2.2.17 takes up the rework.

This is round 2 of the quick route. The method section folds in the posterior scouting report: it computes the posterior over ops and the Bayes ceiling from the op table, with no training, and uses them to pick the three corpus conditions for the control.

Findings

Each rule was to propose a setting for round 3. Only the first was scored.

- The corpus rule (a): no condition passes. The d64-L4 control scored 0.27 at the center against a pass line of 0.49, the other two conditions and the d128-L4 control fell as far short, and none is calibrated. Under the frozen rule, this is the outcome that adds a round.
- Rules (b) to (e) and the two predictions: not reached. They compare anchored arms with a control that has learned about half of what its examples allow, so the comparison would mix the effect of the anchor with unfinished training.

How to read this report

This was a preregistration. The conditions, the five rules with their thresholds, and the two predictions were frozen at commit `49e59eb`, before any run; later edits to them are immaterial or are marked here. The method section on the posterior and the ceiling was computed before training.

Every arm trained and was evaluated, and the eval is in the store, but only rule (a) is scored here. The sections for the other rules keep their frozen text, each with a note that it was not reached. Terms are defined where they first appear and collected in the glossary.

Why this experiment

Ex-2.2.14 anchored an op and every gate passed, but the anchor went to the word that names the op, so the anchored concept was an attribute of one token. The pivot takes op words out of the grammar.

Each line is one context: a few solved examples of one op, written with `?` in place of the op word, then a query under the same op. The op is inferred from the examples, and the anchored concept is "the op in this context is `difference`", which no token names.

This pilot scouts the posterior (see Method below), trains the new-grammar control, smoke-tests anchoring the inferred op, and tries the label variants.

The control is the regression check; a grammar this different needs its own, as ex-2.2.3 did. No model trained on the corpus can know more about the op than the examples say, and the posterior captures that. So instead of fixed accuracy numbers, the task gate is how far the control falls short of the ceiling the posterior sets, with an analytic floor beside it.

The crop policy starts as `whole`, which pulls only labelled lines wholly inside the training window. Ex-2.2.15 found that lines cut short by the window show the first-operand lean (a leak toward the anchor). Its review chose `whole` over `half`, which its rule had set, because a whole line always shows its evidence, which matters more once the evidence is a set of examples, and its discussion asks the pilot to watch the trailing-fragment lean as well.

Glossary

Context

One line of the corpus: *k* examples of one op, then a query. Its op is inferred from the examples.

Example

A solved equation inside a context, `a ? b = y,` : the evidence the op is inferred from.

Query

The last equation of a context, whose answer the model completes. It is always clean.

Replacement op noise

Showing, in some examples, the answer another op would give, at a rate ρ per example. It spreads the posterior over ops, so it grades the stimulus.

Cube noise

Showing, in some examples, a color drawn from the whole grid, at a rate κ. Usually no op produces it, so it removes the evidence of an example without pointing anywhere else.

Posterior on the true op

How strongly the examples of a context point to the op that generated it, computed from the op table under the stochastic rounding and replacement op noise the corpus uses. For `difference` contexts it is the graded stimulus.

Bayes ceiling

The expected exact match on the query of the predictor that holds the posterior over ops and answers with the posterior-weighted answer distribution: the score of a calibrated model that has learned everything the examples say, which is where cross-entropy training aims.

Floor

The same predictor with no evidence: uniform over ops, so its answer distribution is the op-marginal. It is what a model that ignores the examples can score.

Skill score

Where a score sits between the floor (0) and the ceiling (1): (score − floor) / (ceiling − floor).

Expected exact match (EEM)

The probability mass the model puts on the colors the answer of the query can be under the true op, weighted by how often the op gives each, per context, then averaged over held-out contexts. The task metric of every report since ex-2.2.9; the ceiling and the floor are in the same units.

KL divergence

How far one distribution is from another, in nats: $KL(q \| p) = \Sigma_y q(y) \log(q(y) / p(y))$, zero when they agree. Here q is the Bayes predictive on a query and p the model's answer distribution, and the mean over contexts is the model's excess loss over a calibrated one, so 0.05 nats is a twentieth of a nat of loss the model could still shed. The calibration check gives reference values.

Alignment

The cosine between a state and e_1, the anchored axis: 1 when the state points along the axis, 0 when perpendicular. Measured per position and slice.

Lean ($\bar{\alpha}$)

The mean alignment at a role over the held-out contexts of every op: a leak toward the anchor at a site that carries no evidence about the op. The *first-operand lean* is $\bar{\alpha}$ at the first token of a context, at the last block; the *trailing-fragment lean* is $\bar{\alpha}$ over every position, at the last block, of a context cut to start partway through, fed on its own. Both from ex-2.2.15.

Op margin

How far the contexts of the anchored op sit along e_1 beyond the rest: per slice, the mean alignment over the contexts of the anchored op minus the mean over the contexts of all eleven ops, at the role where that gap is largest, averaged over slices. The quantity the anchor term optimizes, so it checks that the pull landed. The statistic of ex-2.2.14, with roles running over the whole context.

Seed band

The smallest difference between two seed means that a comparison resolves: $2\sigma\sqrt{(2/n)}$ at n seeds per arm, with σ the seed standard deviation pooled over the two arms. Rules (b) and (d) use it on expected exact match at the center condition.

Conditions

The pilot trains 14 arms at 3 seeds each, 42 runs. The control runs at the three corpus conditions the method section proposes, `k3-r0.2`, `k3-r0.3`, `k4-r0.3` (three or four examples per context, at a replacement rate of 0.2 or 0.3), at d64-L4, and once more at d128-L4 at the center. The larger control is wider rather than deeper: the part of the task we are least sure a small model can do is the lookup of each example against eleven ops, which we expect to need width more than depth, and keeping four blocks keeps the slice axis the same for every arm. It trains either way, so that a miss under rule (a) does not cost a round, and is scored only if rule (a) needs it. Everything else runs at the center condition, `k3-r0.3`.

The anchored arms all anchor `difference` on e_1 with the recipe from ex-2.2.14 and crop policy `whole`, and differ in what the label covers: the whole context, or one of the four label variants. A hinge arm caps the whole-line pull at an alignment of 0.8, so no state is asked to be all concept. Two pairs add something to the corpus or the model on both the control and the whole-line arm: verification lines, and the newline mask, under which attention does not cross a line break.

Ex-2.2.15 also carried an oracle arm, `knowable`, which pulled a whole labelled line only when its op was in view. Under `whole` every pulled context is wholly in view, so on this grammar that arm is the same as the whole-line arm, and the pilot leaves it out.

arm	group	condition	model	anchored	label	corpus	seeds	what it asks
`control-k3-r0.2`	control	`k3-r0.2`	d64-L4	—	—	—	3	the un-anchored control at each corpus condition
`control-k3-r0.3`	control	`k3-r0.3`	d64-L4	—	—	—	3	the un-anchored control at each corpus condition
`control-k4-r0.3`	control	`k4-r0.3`	d64-L4	—	—	—	3	the un-anchored control at each corpus condition
`control-large`	control	`k3-r0.3`	d128-L4	—	—	—	3	the larger control, trained either way and scored only if rule (a) needs it
`anchor-whole`	label	`k3-r0.3`	d64-L4	yes	`whole`	—	3	the whole context, every slice
`anchor-no-emb`	label	`k3-r0.3`	d64-L4	yes	`no-emb`	—	3	(a) the whole context, the embedding slice left out
`anchor-latter`	label	`k3-r0.3`	d64-L4	yes	`latter`	—	3	(b) the latter half of the context, where the prefix posterior is near its final value
`anchor-prefix`	label	`k3-r0.3`	d64-L4	yes	`prefix`	—	3	(c) a position once the posterior given the tokens before it clears the threshold
`anchor-sampled`	label	`k3-r0.3`	d64-L4	yes	`sampled`	—	3	(d) the whole context, the label drawn with probability equal to the posterior on the op
`anchor-hinge`	hinge	`k3-r0.3`	d64-L4	yes	`whole` + hinge	—	3	the whole-line pull capped by a hinge
`control-verify`	verify	`k3-r0.3`	d64-L4	—	—	verification	3	the control with verification lines
`anchor-verify`	verify	`k3-r0.3`	d64-L4	yes	`whole`	verification	3	the whole-line arm with verification lines
`control-mask`	mask	`k3-r0.3`	d64-L4	—	—	newline mask	3	the control with the newline mask
`anchor-mask`	mask	`k3-r0.3`	d64-L4	yes	`whole`	newline mask	3	the whole-line arm with the newline mask

The 14 arms, 42 runs in all. Every anchored arm anchors `difference` at the center condition; `control-k3-r0.3` is the reference for all of them, and `anchor-whole` is the whole-line arm the label, hinge, verification, and mask arms are compared with. Corpus: what the corpus or attention of the arm adds; verification lines replace 30% of the completion lines.

The labeller. A context of `difference` draws a label with probability 0.02, keyed on the op array beside the corpus rather than on any token; no context of another op is labelled. Under the whole-line label every token of a labelled context is pulled through the pooled anchor term, which asks the context to align somewhere in its span. Variant (a) leaves the embedding slice out of the pull; (b) pulls the latter half of the context; (c) pulls a position once the posterior on `difference` given the examples before it reaches 0.5; (d) makes the draw depend on the evidence: a context of `difference` is labelled with a probability proportional to its posterior on `difference`, with the constant set so that, on average, 2% of them are labelled, as in the other arms. A context whose examples barely fit `difference` is then rarely labelled, and one whose examples all but name it is labelled a little more often than in the other arms.

The seeds. Three per arm, all fresh; no arm is served from the store. Comparisons between arms are between seed means, with the seed band as the resolution.

The corpus rule (a)

The rule. A condition *passes* when the seed-mean held-out expected exact match of the d64-L4 control is at least the ceiling of record less 0.03. Among the passing conditions, the one with the widest spread of the posterior on the true op (its standard deviation across contexts, from the method section; the middle-band share breaks a tie) goes forward. Spread is the criterion because it is a property of the corpus: it says how graded the stimulus is, and round 3 measures the anchor against that grading. The calibration KL (below) is a property of the model, and would favor the condition with the least noise, where there is least to infer and least grading. So the KL is reported beside each condition, and a control that passes without being calibrated is flagged. If none passes, the d128-L4 control at the center is scored on the same line, and goes forward with the center condition if it passes; if it also falls short, the grammar is reworked before anything is anchored. That is the one outcome that adds a round.

The margin is one-sided. The calibrated ceiling is not the most a model can score: a model that sharpens past calibration can reach the hard-EM ceiling, which is higher (the method section has both). So a control above the ceiling passes, and the calibration check below says whether it got there by sharpening, which is worth knowing before round 3 reads its answer distribution. The bounds of record at each condition, and the score the control has to reach:

condition	ceiling (EEM) ↑	floor	room	pass line (ceiling − margin)	margin / room
`k3-r0.2`	0.594	0.108	0.485	0.564	0.06
`k3-r0.3`	0.517	0.109	0.409	0.487	0.07
`k4-r0.3`	0.577	0.109	0.468	0.547	0.06

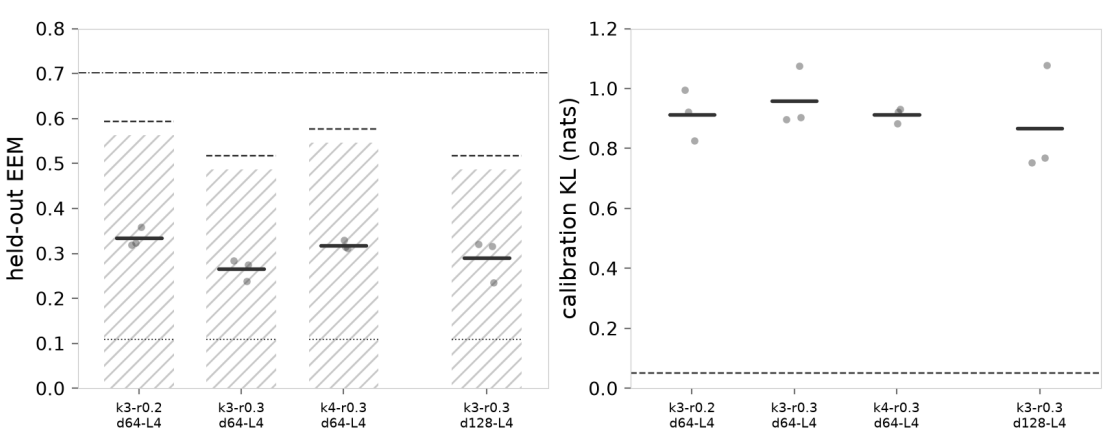
The bounds of record for rule (a), at the cube-noise rate of the corpus, $\kappa = 0.02$: the calibrated Bayes ceiling, the floor, the room between them, the score a control has to reach, and the margin as a share of the room (the pass line is a skill score of one minus this).

At the center the margin is about 7% of the room between floor and ceiling. For scale, a predictor that ignores one of the three examples and is otherwise Bayes-optimal gives up 0.097 there, so a control that passes has learned to use every example, near enough.

What we expect. The d64-L4 control passes at every condition, and `k3-r0.3` goes forward. The pivot expects the model to be enough because there are no variables to track; the inference is a lookup over eleven ops per example and a pool over examples. A control that passes at `k3-r0.2` and misses at the center would say the noise, not the grammar, is what costs it, and the fallback within the same block size is already in the table.

The calibration check. The check is the mean over held-out contexts of the KL divergence from the Bayes predictive q (the posterior-weighted answer distribution on the query) to the model's answer distribution p at the query `=`, in nats. Averaged over contexts, it is the model's cross-entropy on the answer less the Bayes-optimal cross-entropy: the loss the model could still shed. A control is *calibrated* at a condition when it is under 0.05 nats. The method section gives the scale: a tenth of the floor mixed into the Bayes answer costs about 0.043 at the center, ignoring one example 0.29, and committing to the most probable op 0.73. We expect the control to be calibrated at every condition it passes; a control that passes and is not calibrated has sharpened, and the direction of the miss (the KL grows fastest where the model puts no mass on an answer q allows, so a sharpened model shows it more than a hedging one) is reported beside it.

The result. No condition passes, and the d128-L4 control misses too:



The controls against the ceiling. *One column per control, the d128-L4 control at the right after a gap.* ***Left:*** *held-out expected exact match, seeds as faded dots and the seed mean as a bar. Per column, the ceiling of record is the dashed segment, the floor the dotted one, and the region below the pass line (the ceiling less 0.03) is hatched. The dash-dot rule is the ceiling for a model told the op (0.702).* ***Right:*** *the calibration KL in nats, with the threshold of 0.05 as a dashed rule.*

control	model	EEM per seed	seed mean ↑	pass line	skill score	spread	KL ↓
`k3-r0.2`	d64-L4	0.359, 0.324, 0.319	0.334	0.564	0.46	0.28	0.91
`k3-r0.3`	d64-L4	0.284, 0.239, 0.276	0.266	0.487	0.38	0.33	0.96
`k4-r0.3`	d64-L4	0.314, 0.311, 0.329	0.318	0.547	0.45	0.29	0.91
`k3-r0.3`	d128-L4	0.235, 0.316, 0.321	0.291	0.487	0.45	0.33	0.87

Rule (a), scored. Held-out expected exact match over all contexts per seed and its seed mean, the pass line of the condition, the skill score of the seed mean, the spread of the posterior on the true op (the criterion among passing conditions), and the seed-mean calibration KL in nats. No row passes.

The shortfall is large and much the same everywhere. Every control has a skill score between 0.38 and 0.46, a little under halfway from the floor to the ceiling, and every seed sits far below its pass line. Width did not help: the d128-L4 control at the center scored 0.291, against 0.266 for the d64-L4 control, a difference inside the spread of its own seeds (0.235 to 0.321). The calibration KL is 0.75 to 1.08 nats across the seeds, many times the threshold of 0.05.

So the grammar is reworked before anything is anchored, as the rule says. The miss alone does not say whether the grammar or the training recipe is the limit; the discussion has what came next.

The label rule (b)

The rule. The whole-line label stays the primary unless a variant *clears the task gate*: its seed-mean held-out expected exact match at the center, over all contexts, exceeds that of the whole-line arm by more than the seed band, while it *holds the anchor*: its op margin is at least 90% of the margin of the whole-line arm. If more than one variant qualifies, the one with the higher op margin goes forward. Variant (d) trains the grading that the evidence section reads, so it is reported on the same statistics and not promoted.

The op margin is the one ex-2.2.14 used, with the contexts of the anchored op as the labelled group and roles running over the whole context (the glossary has it). The variants narrow the pull, so their margins can only be lower by construction where the narrowed positions carried it; the bar asks that most of it survives.

What we expect. No variant clears the gate. Ex-2.2.14 found that the label share barely moves the margin and that the task is untouched at the rate of the primary, and the variants change less than that. What they may change is where the alignment sits: (a) should leave the embeddings clean, (b) and (c) should put less alignment on the first example, and (c) leaves the middle-band contexts unlabelled. Those are read in the alignment figures of the evidence section; this rule scores the task and the margin only.

Not reached. Rule (a) did not pass, so this is not scored. The arms trained, and their eval is in the store.

The hinge rule (c)

The rule. The uncapped whole-line arm *saturates* at the query `?` when the seed-mean alignment at that position, on the held-out contexts of `difference` at the last block, is at least 0.9. If it does, the hinge arm goes forward in its place, provided the hinge arm holds the anchor (90% of the whole-line margin) and it is within the seed band of the whole-line arm on the task. If the whole-line arm does not saturate, it stays, and the hinge arm is reported.

Saturation is the dose collapse ex-2.2.14 found at the op word: a state that is all concept has no partial dose, because projecting e_1 out of it leaves a remainder too small to mean anything. The query `?` is a constant token with nothing else to hold, so the pooled pull may put the alignment of the whole context there. The hinge counts an alignment of 0.8 as full: in the anchor term only, a position pays $1 - \cos/0.8$ in place of $1 - \cos$, and nothing above the cap. A sharp corner at the cap would switch the pull off there while the anti-subspace term keeps pushing the state back down, so a state near the cap could flip between pulled and not pulled from step to step. So the corner is rounded with a softplus, a smooth version of max(0, x), of width 0.05, and the pull fades out over a narrow band around the cap, which gives the two terms a smooth point to balance at. We expect the pulled states to settle near 0.8, keeping a part off the axis for the projection to land on. Measurements use the raw cosine.

What we expect. We do not predict either way. The pooled term concentrates the pull where alignment comes cheapest, which argues for saturation; the anti-subspace term penalizes the squared alignment of every state, which argues against it, and *red* under the same recipe reached about 0.5 at its own token. What we do expect is that the query `?` carries more alignment than any example position on the whole-line arm, since it is the first position at which the whole context has been seen and its state is free.

Not reached. Rule (a) did not pass, so this is not scored. The arms trained, and their eval is in the store.

The verification rule (d)

The rule. Verification lines stay in the corpus from round 3 on if they *leave completion unchanged*: the seed-mean held-out expected exact match on completion contexts is within the seed band of the arm without them, on both pairs, `control-verify` against `control-k3-r0.3` and `anchor-verify` against `anchor-whole`. Both pairs have to hold; a corpus that costs the anchored arm and not the control would say the anchor and the second task compete.

A verification line replaces the completion of 30% of contexts with a candidate answer, a marker, and a `TRUE` or `FALSE` verdict (the pivot); the loss is masked on a `FALSE` candidate (the answer token only). The verification accuracy itself (the share of verdicts right, held out) is reported with no gate, as the first number D2.3 gets; so is the op margin of `anchor-verify` against the margin of the whole-line arm.

What we expect. Completion is unchanged on both pairs. The verification arms see 70% of the completion lines the others do, over the same steps, and the calibration look in ex-2.2.9 found that the task is limited by distinct lines rather than passes over them; so the cost, if there is one, shows on the ops that round the most. A miss on the control pair would send the rate down before round 3 rather than take verification out.

Not reached. Rule (a) did not pass, so this is not scored. The arms trained, and their eval is in the store.

The operator rule (e)

The rule. A scoring-only suppression pass runs on the checkpoints of the pilot itself, on the whole-line arm, the hinge arm, and the control at the center: the projection along its dose axis (γ in 0.25, 0.5, 0.75, 1: the fraction of the component on e_1 removed), the repulsion along its dose axis (a landing at 0.25, 0, for states above an alignment of 0.5), and the reflection (the projection at $\gamma = 2$, one dose, as a reference). Each acts at every slice, at three sites: the query `?`, the query `=`, and every position. The rule is scored on the every-position edit, on the anchored arm that rule (c) sends forward: the whole-line arm, or the hinge arm if the whole-line arm saturates. The other anchored arm is reported beside it. An operator *grades with dose* when the seed-mean drop in expected exact match on the held-out `difference` contexts is non-decreasing along its dose axis and reaches, at full dose, at least 50% of the way from the clean score to the target null; it is *selective* when on each of the other ten ops the seed-mean drop is at most 0.02 at every dose, with the control under the same edit as the reference. The operator that does both goes forward with its dose axis; if both do, the projection, which is the simpler; if neither does, the operator for round 3 stays open, and the pilot reports why.

The target null is the answer a model that has lost `difference` and nothing else would give: the posterior-weighted answer distribution with `difference` removed and the rest renormalized. Its expected exact match on a context is the target the full-dose damage is measured against, per context, from the posterior of the method section.

What we expect. The projection grades if the query `?` does not saturate, and collapses (all of its damage in the last step of γ) if it does; the repulsion grades either way, since its dose is where the state lands rather than how much is removed. Selectivity should hold for both at the query sites, which the other ops' contexts share only as syntax, and is the open question at every position, where the edit touches states that carry the other ops. The two query sites are the first bypass measurement: an edit at `?` that does less than the same edit at `=` says the model reads the op from the examples rather than from `?`.

Not reached. Rule (a) did not pass, so this is not scored. The arms trained, and their eval is in the store.

The anchor follows the evidence

What we expect. On the whole-line arm, the alignment at the query `=` on a held-out `difference` context rises with the posterior on `difference` across the middle band (0.5 to 0.95): binned by posterior, the seed-mean alignment at the last block is higher in each bin than the one below it. That is the graded stimulus, the part redness played for *red*, and it is a result the pull did not train for: the label is binary on the true op, so a context whose examples half-fit `difference` was pulled as hard as one that names it. A shortcut, such as a characteristic answer color, would likely not follow the posterior.

Variant (c) leaves the middle-band contexts unlabelled, so grading there under (c) is the cleaner version of the same result; variant (d) trains it and is the comparison. Alignment against the posterior on the *other* ops' contexts (where the posterior on `difference` is low but not zero) is reported beside it, since a graded response should be low and flat there.

Not reached. Rule (a) did not pass, so this is not scored. The arms trained, and their eval is in the store.

The leans under `whole`

What we expect. On the whole-line arm, both leans stay within the seed band of the control. Ex-2.2.15 found that under `all`, lines cut short by the window taught the model to lean toward e_1 at the first operand of every line, and more so on a trailing fragment shown alone; under `whole` both leans sat inside the band of the control. This grammar gives a cut context more to lean with, since a window cuts more contexts than lines and a fragment of a context still looks like a context, so the lean is worth reading again. The first-operand lean is $\bar{\alpha}$ at the first token of the context at the last block; the trailing-fragment lean is $\bar{\alpha}$ over every position, at the last block, of a context cut to start at each example boundary, fed on its own; both over the held-out contexts of every op, with the mask arms measured with the mask on.

Not reached. Rule (a) did not pass, so this is not scored. The arms trained, and their eval is in the store.

Discussion

The pilot expected the d64-L4 control to reach its ceiling at every condition, on the reasoning that inferring the op is a lookup over eleven ops per example and a pool over examples. It got a little under halfway instead, at every condition and at twice the width. Validation loss was still falling at the end of training, so the first question after the pilot was whether the recipe or the task was the limit.

Ex-2.2.17 is a scout on that question, on the center control. A newline mask, a lower peak learning rate, and eight times the steps lift it to about 0.45, still short of the pass line, and no schedule, length, width, or depth moves it further. Its scoring places the remaining gap on contexts whose examples settle the op, and on one op, `hsvmix`, that the model computes poorly. Its backlog items on the op set (similar ops, `hsvmix`) are the grammar side of the rework.

None of rules (b) to (e) was tested here, so they can carry over to the next pilot unchanged.

Method

The *posterior* error ops

Each example is a pair of *typechecked* and *un-erased*, and *order-dependent* counting the answer is zero even when it can't claim to be the *likelihood* of a chosen answer *x* under an *op* is the probability of recalling the result of a unit test *op* on the *x* posterior $P_{\mathcal{N}}(x, \theta)$ (Jensen, 2014 - *Acta Mater.*).

The model is *tuned* and *hyperparameters* by *gradient* to where the *raw* value *lies* between grid *cells*. The *posterior* uses the *counting*.

Replacement *op* was *erases* the *likelihood* *x*, which *lies* up a *sum* example shows a *bar* from the *distribution* of another *op* instead, the *error op* uniform such the *op* is

$$E_{\mathcal{N}}(x, \theta) = (x - \theta)^2 + \frac{1}{\sum_{i=1}^{n} \mathbb{1}(x_i = x)} \cdot \frac{\partial \mathcal{L}}{\partial \theta}$$

where $E_{\mathcal{N}}$ is the *likelihood* of the *lowest answer* under *op* in the *Poisson* $\mathcal{L}_{\text{Poisson}}(\lambda) = \lambda_i$ the *mean* *raw* and *x* is the *raw* *cells* *lies* *raw* *cells* *raw*. The *posterior* over the *answer* *op* is the *product* of the *likelihood*. *likelihood* under a *uniform* prior. In the *posterior* is *model* *answer* on *key* *cross* *op* *work* at *test*. It also *tests* *gradient* *descent* *op* *raw* *cells* *raw* *cells* *raw* *cells